

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

November 2013

CL 210 Hydrology & Hydraulics

Date: 14.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1(a) What are the factors affecting evaporation? Suggest the measures for reducing evaporation. [04]

(b) Define: Infiltration capacity. Write Horton's equation for the determination of infiltration capacity and explain each term in it. [03]

Q - 2(a) Derive an expression for the steady state discharge of well fully penetrating into a confined aquifer. [06]

(b) Distinguish between the following: [08]

- (i) Confined aquifer and unconfined aquifer (ii) Shallow well and deep well.

A 30 cm diameter well fully penetrates a confined aquifer of thickness 35m. When the well is pumped at a rate of 120 lit/min, the steady state drawdown in observation wells located at 10 m and 100 m radial distance from the pumping well are found to be 5.5 m and 0.1 m respectively. Calculate the coefficient of permeability of the aquifer.

OR

Q - 2 (a) A fully penetrating well of dia. 0.3m is located in an unconfined aquifer of saturated depth of 45m. If the drawdown in the well is 15m for the discharge of 1200 m³/day, and the radius of influence is 300m, compute hydraulic conductivity. [05]

(b) Enlist various sub-surface techniques of ground water investigation. Discuss any two techniques in detail. [05]

(c) What are the assumptions made in the analysis of steady radial flow into well. [04]

Q - 3 (a) Explain 'Artificial recharge of groundwater'. Enlist different methods of groundwater recharge and explain, with a neat sketch, one of them. [07]

(b) What preventive measures shall be taken to control the sea water intrusion? [04]

- (c) Discuss the phenomenon of "Upconing of saline water". [03]

OR

- Q - 3 (a) Discuss various ground water recharge estimation techniques briefly. [07]
- (b) Explain Ghyben-Herzberg theory for salt water intrusion, with a neat sketch. Derive the Ghyben-Herzberg relation for salt water-fresh water interface. [07]

SECTION - II

- Q - 4 (a) What are the advantages and disadvantages of Roof Rain Water Harvesting System? [04]
- (b) Which are the factors affecting ground water pollution? [03]
- Q - 5 (a) Discuss briefly the pollution from wells and salt water. [03]
- (b) Normal annual rainfall at stations A, B, C and D are 60.97, 65.28, 78.12 & 90.14 cm respectively. In a the year 1985, the station D was inoperative and the annual precipitation recorded at station A, B, C are 81.12, 68.15 and 75.50 cm respectively. Estimate the annual rainfall at station D at that year. [05]
- (c) Discuss Float type recording raingauges in detail with neat figures. [06]

OR

- Q - 5 (a) What is ground water pollution? Discuss the pollution mechanism briefly. [03]
- (b) Discuss various factors affecting runoff in detail. [05]
- (c) A storm over a catchment of area 5.0 km^2 had duration of 14 hr. The mass curve of rainfall of the storm is as follows: [06]

Time from start of storm (hr)	0	2	4	6	8	10	12	14
Accumulated rainfall (cm)	0	0.6	2.8	5.2	6.6	7.5	9.2	9.6

If the ϕ - index for the catchment is 0.4 cm/hr , determine the effective rainfall hyetograph and the volume of direct runoff from the catchment due to the storm.

- Q - 6 (a) Define "Water Conservation". [02]
- (b) Distinguish with sketches, the difference between an unconfined and a confined aquifer. [06]

- (c) In a certain Place of Madhya Pradesh, the average thickness of the aquifer of the confined aquifer is 20 m and extends over an area of 720 km². The piezometric surface fluctuates annually from 22 to 15 m above the top of aquifer. What ground water can be expected annually, if a storage coefficient is 0.00072? Assuming, an average well yield of 25 m³/hr and about 150 days of pumping in a year, how many wells can be drilled in the area? [06]

OR

- Q - 6 (a) Enlist components of Roof Rain Water Harvesting System. [02]
- (b) Define: Aquifer, Aquitard, Transmissibility, Storage Coefficient, Porosity, and Permeability. [06]
- (c) It was observed in a field test that 2 hr 50 min was required for a tracer to travel from one well to another 18 m apart and the difference in their water surface elevations was 0.75 m. Samples of the aquifer between the wells indicated a porosity of 10%. Determine the permeability of the aquifer, seepage velocity and the Reynolds number for the flow, assuming an average grain size of 1 mm and ν at 27°C = 0.009 Stoke. [06]
